

XG Boost

High level description:

- ① Base prediction
- ② Compute gradients/errors
- ③ Build tree with best splits
- ④ Add scaled tree to predictions
- ⑤ Repeat

Building trees

XG Boost minimises the loss function and two other terms:

$$\underbrace{\sum_{i=1}^n L(y_i, \hat{y}_i)}_{\text{Loss function}} + \underbrace{\gamma T}_{\text{terminal nodes}} + \underbrace{\frac{1}{2} \lambda O_{\text{value}}^2}_{\text{Regularization}}$$

Loss function is different for Regression vs Classification:

Regression: $L(y_i, \hat{y}_i) = \frac{1}{2} (y_i - \hat{y}_i)^2$

Classification: $L(y_i, \hat{y}_i) = -[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$

Second order Taylor approximation for loss function

$$\begin{aligned} L(y_i, \hat{y}_i + O_{\text{value}}) &\approx L(y_i, \hat{y}_i) + \left[\frac{d}{d\hat{y}_i} L(y_i, \hat{y}_i) \right] O_{\text{value}} + \frac{1}{2} \left[\frac{d^2}{d\hat{y}_i^2} L(y_i, \hat{y}_i) \right] O_{\text{value}}^2 \\ &\approx L(y_i, \hat{y}_i) + \underbrace{g_i}_{\text{gradients}} O_{\text{value}} + \underbrace{h_i}_{\text{hessians}} O_{\text{value}}^2 \end{aligned}$$

Minimising loss function

With Taylor approx. we get:

$$\sum_{i=1}^n L(y_i, \hat{y}_i) + g_i O_{\text{value}} + h_i O_{\text{value}}^2 + \frac{1}{2} \lambda O_{\text{value}}^2$$

$L(y_i, \hat{y}_i)$ is constant to O_{value} so we can remove:

$$\sum_{i=1}^n g_i O_{\text{value}} + \frac{1}{2} \left[\left(\sum_{i=1}^n h_i \right) + \lambda \right] O_{\text{value}}^2$$

Derivate and solve for O to find minimum:

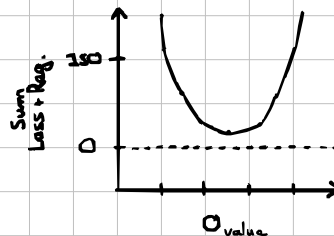
$$\frac{d}{dO_{\text{value}}} \sum_{i=1}^n g_i O_{\text{value}} + \frac{1}{2} \left[\left(\sum_{i=1}^n h_i \right) + \lambda \right] O_{\text{value}}^2 = 0$$

$$\Leftrightarrow \sum_{i=1}^n g_i + \left(\sum_{i=1}^n h_i + \lambda \right) O_{\text{value}} = 0 \Leftrightarrow O_{\text{value}} = \frac{-\sum_{i=1}^n g_i}{\sum_{i=1}^n h_i + \lambda} \quad (\text{optimal } O_{\text{value}} \text{ for the leaf})$$

Compute Similarity score with optimal value:

$$\text{Score} = \frac{(\sum_{i=1}^n g_i)^2}{\sum_{i=1}^n h_i + \lambda}$$

$$\text{Total gain for split: } \frac{1}{2} (\text{Score}_L + \text{Score}_R - \text{Score}_p) - \gamma$$



- ① Gradients inside node strongly agree in direction: $|\sum g_i|$ large, score large
- ② Hessians large, curvature of gradients is high so less confidence over a large move
↳ reduce score

Parameters

- learning_rate, step size per tree, lower = more trees, but more robust
- max_depth, max depth of each tree, higher = more complex tree, overfitting
- subsample, row sampling, adds randomness, reduces overfitting
- colsample_bytree, feature sampling per tree, same idea on columns / features
- n_estimators, number of trees
- min_child_weight, min. samples in a leaf